

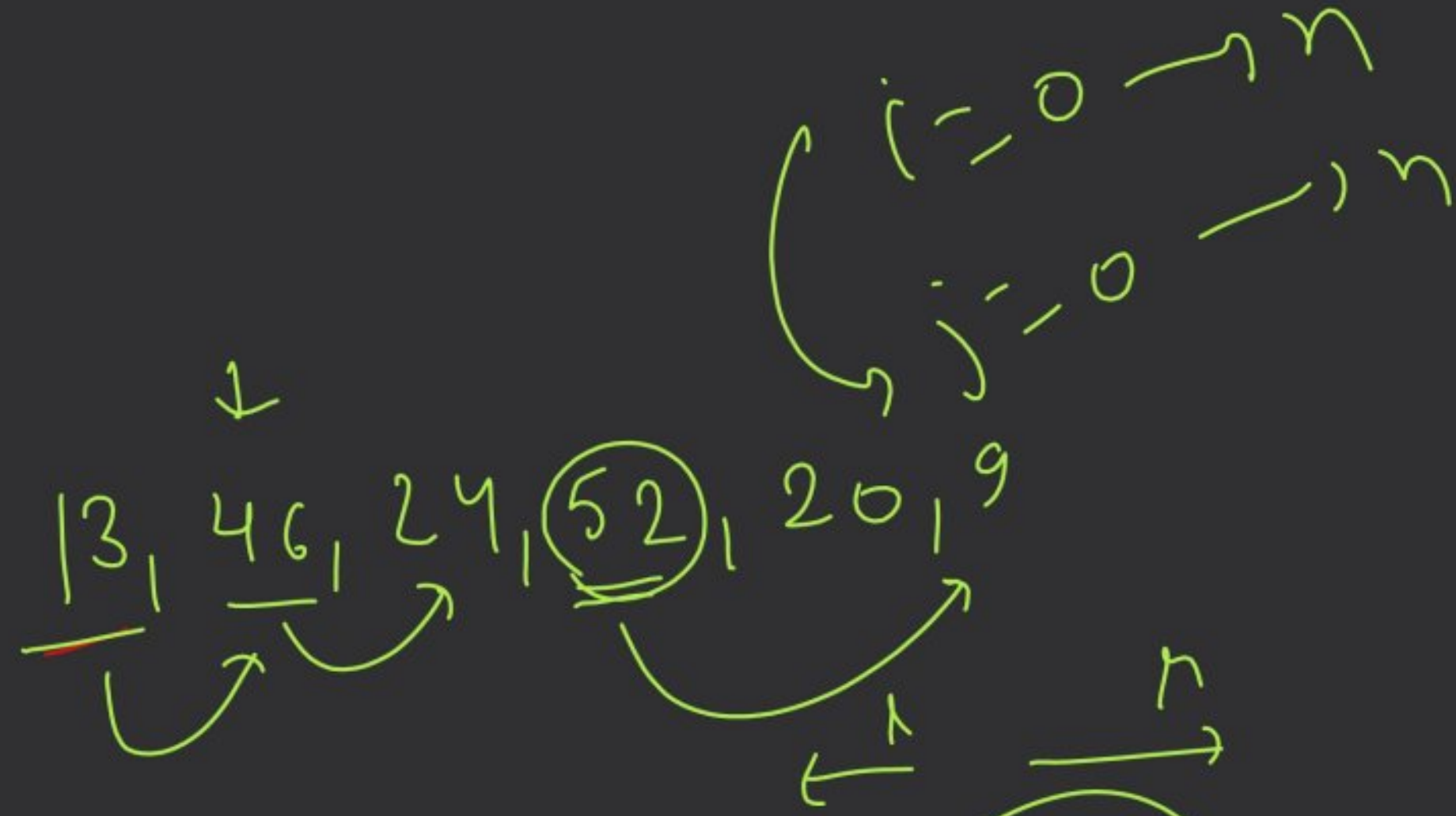
Bubble Sort :- push maximum no.
to the last

(What is sorting)

[13, 46, 24, 52, 20, 9]

↓
[9, 13, 20, 24, 46, 52]

(Merge & Quick)



$(46 > 24)$ $13, 24, 46, 52, 20, 9$

$13, 24, 46, 20, 52, 9$ $\rightarrow 13, 24, 46, 20, 9, 52$

$i=1$
 $13, 24, 46, 20, 9, 52$
 $\uparrow \quad \uparrow$
 $20 \quad 46$
 $13 \quad 24 \quad 20 \quad 9 \quad 46 \quad 52$

$S-II: - 13 \quad 20 \quad 24 \quad 9 \quad 46 \quad 52$

$S-IV: - (13 \quad 9) \quad 20 \quad 24 \quad 46 \quad 52$
 $(9 \quad 13 \quad 20 \quad 24 \quad 46 \quad 52)$

$(5, 3, 1, 4, 2)$
 $\downarrow$
 $\rightarrow 3, 5, 1, 4, 2$
 $\rightarrow 3, 1, 5, 4, 2$
 $\rightarrow 3, 1, 4, 5, 2$
 $\rightarrow 3, 1, 4, 2, 5$

$(1, 3, 4, 2, 5)$
 $(1, 3, 2, 4, 5)$

$i=0, \quad j=0 \rightarrow n-1-0$

$i=1, \quad j=0 \rightarrow \underline{n-1-1}$

$i=2, \quad j=0 \rightarrow \underline{n-1-2}$

$i=3, \quad j=0 \rightarrow \underline{n-1-3}$

```
for (int i=0 ; i < n-1 ; i++) {  
    for (int j=0 ; j < n-1-i ; j++) {  
        if ( arr[j] > arr[j+1] ) {
```

// swap

—

←

}

Selection Sort :-

(select min & swap)

for (int i=0 → i<n-1) {

 min = i;

 for (j = i+1 → j<n) {

 if (arr[j] < arr[min]) {

 min = j;

 } swap(arr[min], arr[i]);

[13, 46, 24, 52, 9]

[9, 46, 24, 52, 13]

[9, 13, 24, 52, 46]

[9, 13, 24, 46, 52]

[5, 3, 1, 4, 2]

(i=0)

[1, 3, 5, 4, 2]

[1, 2, 5, 4, 3]

[1, 2, 3, 4, 5]

[1, 2, 3, 4, 5]

i=0

5, 3, 1, 4, 2

i ↑ j=1

min = 0, 1, 2, 3, 4

arr[j] < arr[min]

Insertion Sort :-

[5, 3, 1, 4, 2]

(key = 3)

(i = 1)

shift 5 $\Rightarrow$ [3, 5, 1, 4, 2]

i = 2, key = 1, [1, 3, 5, 4, 2]

i = 3, key = 4, [1, 3, 4, 5, 2]

i = 4, key = 2, [1, 2, 3, 4, 5]

[14, 9, 15, 12, 6, 8, 13]

(i = 1)

[1, 3, 4, 5, 2]

(i = 4, key = 2)

(arr[j] = 5 > 2)

arr[j+1] = key

[1, 3, 4, 5, 5]

[1, 3, 4, 4, 5] $\rightarrow$ [1, 2, 3, 4, 5]

```

for(int i=1 → n){
    int key = arr[i];
    int j = i-1;
    while(j >= 0 && arr[j] > key){
        arr[j+1] = arr[j];
        j--;
    }
    arr[j+1] = key; ←
}

```

[5, 3, 1, 4, 2]
 ↑
 i
 (key = 3)
 [5, 3, 1, 4, 2]
 ↑
 j = 0
 [5, 1, 1, 4, 2]
 (j = -1)
 (key = 1)
 [3, 5, 5, 4, 2]
 ↓
 [2, 3, 5, 4, 2]
 ↑
 1

/ key = 4
 j = i-1
 [1, 3, 5, 4, 2]
 (5 > 4)
 j
 [1, 3, 5, 5, 2]
 3 > 4 (false)
 [1, 3, 4, 5, 2]
 (i=4, key=2)

$[1, 3, 4, 5, 2]$

$(j=3)$

$\downarrow j$

$key=2$

$[1, 3, 4, 5, 5]$

$\downarrow j$

$[1, 3, 4, 4, 5]$

$\downarrow j$

$[1, \underline{3}, 3, 4, 5]$

$3 > 2 \checkmark$

$1 > 2 \times$

$[1, 2, 3, 4, 5]$

—

—

Pivot Index :-

nums = [1, 7, 3, 6, 5, 6]

0 1 2 3 4 5
↔ ↓ ↔
" " "

pivot
index

(is a index from sum of all left
element == sum of right element)

[-1, 0, 0, 0]

10 → pivot index



[1, 7, 3, 6, 5, 6]

(total sum = 28)

$i=0$, left sum = 0
right sum = 27

$i=1$, left sum = 1

right sum = 20 = total sum - current - left sum

$i=2$, left sum = 8

right sum = 17



$i=3$

left sum = 11

right sum = 28 - 6 - 11 = 11

```
int totalSum = 0;
```

```
// totalSum
```

```
int leftSum = 0;
```

```
for (int i = 0; i < n; i++) {
```

```
    int rs = totalSum - nums[i] - leftSum;
```

```
    if (ls == rs) {
```

```
        cout(i);
```

```
        break;
```

```
    }
```

```
    leftSum = leftSum + nums[i];
```

①

← (1, 7, 3, 6, 5, 4) →

(i=0)

ls = 0

rs = 27

// ls initialised at the end

// break must be there

// formula is important

[2, 1, -1]
↑

i=0
(ls=0, rs=0)

static int pivotIndex(int[] nums) {
// total sum

int ls = 0;

for (int i = 0; i < n; i++) {

int curEl = nums[i];

int rs = totalSum - curEl - ls;

if (ls == rs) { (2, 1, -1)

return i;

}

ls = ls + nums[i];

} ←
return -1;

FileEditSelectionView

LAPTOP-HR674551

08112024

selectionSort.javaBubbleSort.javaXInsertionSort.java

BubbleSort.java > BubbleSort > main(String[])

```
1 public class BubbleSort {
2
3     public static void bubbleSort(int[] arr) {
4
5         for(int i = 0; i < arr.length - 1; i++) {
6
7             for(int j = 0; j < arr.length - i - 1; j++) {
8
9                 if(arr[j] > arr[j + 1]) {
10
11                     // swap
12                     int temp = arr[j];
13                     arr[j] = arr[j + 1];
14                     arr[j + 1] = temp;
15                 }
16             }
17         }
18     }
19
20     public static void main(String[] args) {
21
22         int[] arr = {5, 3, 1, 4, 2};
23     }
```

Run | Debug

Ln 29, Col 10 Spaces: 4 UTF-8 CRLF () Java Go Live Prettier

Handwritten notes in red:

- `int h` (next to line 3)
- `ind` (next to line 5)
- `(swap)` (next to the swap block)
- `10-14-1` (with arrows pointing to the inner loop and swap block)
- `[ - , - , - , - ]` (array representation)
- `(7, 2, 5, 3, 1, 0)` (array representation)



Problem List



1:1 Sessions with me



81



Premium

</> Code



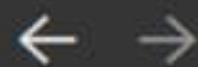
Java Auto



```
1 class Solution {
2     public int pivotIndex(int[] nums) {
3         int n = nums.length;
4         int totalSum = 0;
5
6         for(int num : nums){
7             totalSum += num;
8         }
9
10        int leftSum = 0;
11
12        for(int i = 0; i < n; i++){
13            int rightSum = totalSum - nums[i] - leftSum;
14
15            if(leftSum == rightSum) return i;
16
17            leftSum += nums[i];
18        }
19        return -1;
20    }
21 }
```

Problem 724 of leetcode





Q Codes



selectionSort.java

BubbleSort.java

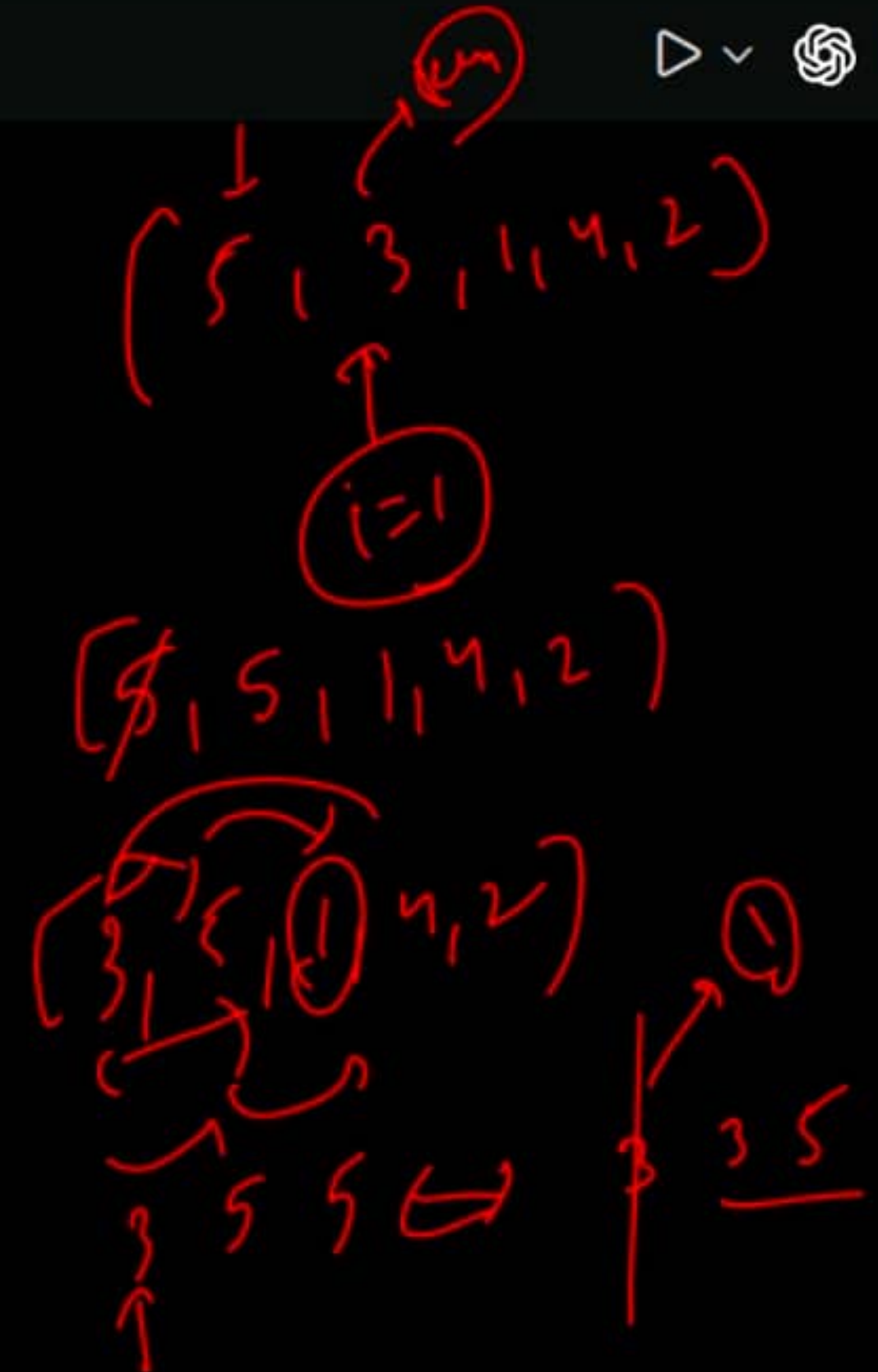
InsertionSort.java X



InsertionSort.java > InsertionSort > main(String[])

```
1 public class InsertionSort {
2
3     public static void insertionSort(int[] arr) {
4
5         for(int i = 1; i < arr.length; i++) {
6
7             int key = arr[i];
8             int j = i - 1;
9
10            // shift elements
11            while(j >= 0 && arr[j] > key) {
12                arr[j + 1] = arr[j];
13                j--;
14            }
15
16            arr[j + 1] = key;
17        }
18    }
19 }
```

Left side



Run Testcases



Ln 30, Col 6

Spaces: 4

UTF-8

CRLF

{ } Java

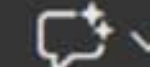


Go Live





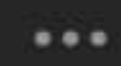
Q Codes



selectionSort.java X

BubbleSort.java

InsertionSort.java



selectionSort.java

```
1 public class SelectionSort {
2
3     public static void selectionSort(int[] arr) {
4
5         for(int i = 0; i < arr.length - 1; i++) {
6
7             int minIndex = i;
8
9             for(int j = i + 1; j < arr.length; j++) {
10                 if(arr[j] < arr[minIndex]) {
11                     minIndex = j;
12                 }
13             }
14
15             // swap
16             int temp = arr[i];
17             arr[i] = arr[minIndex];
18             arr[minIndex] = temp;
19         }
20     }
21 }
```



Run Testcases



0 0



Java: Ready



Ln 18, Col 34

Spaces: 4

UTF-8

CRLF

{ } Java



Go Live



Prettier

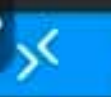




selectionSort.java

BubbleSort.java X

InsertionSort.java



BubbleSort.java > BubbleSort > main(String[])

```
1 public class BubbleSort {
2
3     public static void bubbleSort(int[] arr) {
4
5         for(int i = 0; i < arr.length - 1; i++) {
6
7             for(int j = 0; j < arr.length - i - 1; j++) {
8
9                 if(arr[j] > arr[j + 1]) {
10
11                     // swap
12                     int temp = arr[j];
13                     arr[j] = arr[j + 1];
14                     arr[j + 1] = temp;
15
16                 }
17             }
18         }
19     }
```

Run | Debug

```
20 public static void main(String[] args) {
21
22     int[] arr = {5, 3, 1, 4, 2};
23 }
```

int n

(swap)

10-1-n-1

[7, 2, 5, 3, 1, 0]



Run Testcases



Java: Ready



Ln 29, Col 10

Spaces: 4

UTF-8

CRLF

{ } Java



Go Live

